

Repeated Games Theory

Repeated Games Theory studies situations where the same players interact multiple times instead of only once. Because players know they will meet again, they may cooperate, punish cheating, reward good behavior, and build trust over time.

Unlike a one-shot game, where players make decisions only once, repeated games allow today's actions to influence tomorrow's outcomes.

Definition

A Repeated Game is a strategic game in which the same stage game is played repeatedly by the same players over several periods, allowing them to condition current actions on past behavior.

Key Idea

The possibility of future interactions changes behavior.

- In a one-time interaction:
 - Players often act selfishly.
- In repeated interactions:
 - Cooperation becomes more likely because players fear future punishment or value future rewards.

This idea is often summarized as:

"The future disciplines the present."

Components of Repeated Games

1. Players

The same players interact repeatedly.

Example:

- Two firms
- Two countries
- Employer and employee

2. Stage Game

The basic game played each round.

Example:

- Prisoner's Dilemma
- Pricing Game

- Bargaining Game

3. History

Players remember previous actions.

4. Strategy

A player's current action depends on previous rounds.

5. Payoff

Total payoff equals the sum (or discounted sum) of payoffs across all rounds.

1. Finite Repeated Games

The game has a fixed number of rounds.

Example:

- 10 chess matches
- Five negotiations
- Three business meetings

Characteristics

- End is known.
- Players may stop cooperating near the end.
- Backward induction often predicts defection in every round for games like the Prisoner's Dilemma.

2. Infinite Repeated Games

The game has no known ending.

Examples:

- Long-term business competition
- International diplomacy
- Marriage
- Employer–employee relationships

Characteristics

- Cooperation is much easier.
- Future consequences matter.
- Reputation becomes valuable.

Common Strategies in Repeated Games

1. Always Cooperate

Always cooperate regardless of the opponent's actions.

Advantages:

- Builds trust.

Disadvantages:

- Easily exploited.

2. Always Defect

Always act selfishly.

Advantages:

- Gains immediate benefit.

Disadvantages:

- Destroys long-term cooperation.

3. Tit-for-Tat

One of the most famous strategies.

Rules:

1. Start by cooperating.
2. Copy the opponent's previous move.

Advantages:

- Nice
- Forgiving

4. Grim Trigger

Cooperate until the opponent cheats once.

After one betrayal:

Defect forever.

This strategy strongly discourages cheating.

5. Generous Tit-for-Tat

Mostly copies the opponent but occasionally forgives defections to restore cooperation, especially when mistakes may occur

Discount Factor (δ)

Repeated games often use a **discount factor (δ)** to represent how much players value future payoffs.

- δ close to **1** → Players care a lot about the future, making cooperation more attractive.
- δ close to **0** → Players focus on immediate rewards, making defection more tempting

Importance of Repeated Games

Repeated games explain why cooperation can emerge even among self-interested players.

Applications include:

- Business competition
- International relations
- Environmental agreements
- Labor relations
- Online marketplaces
- Supply chain partnerships

Advantages

- Encourages long-term cooperation.
- Builds trust and reputation.
- Explains stable relationships.
- Reflects many real-world interactions.
- Models punishment and forgiveness.

Limitations

- Requires repeated interactions.
- Assumes players remember past behavior.
- Cooperation may collapse if the game is about to end.
- Complex strategies can be difficult to analyze.
- Uncertainty and communication errors may affect outcomes.

Applications

- Economics (pricing, collusion, long-term contracts)
- Political Science (international treaties, diplomacy)
- Business (customer relationships, supplier partnerships)
- Sociology (social norms and reciprocity)
- Artificial Intelligence (multi-agent systems)
- Environmental Economics (climate agreements)
- Cybersecurity (trust and repeated interactions in networks)

Difference Between One-Shot and Repeated Games

<u>Feature</u>	<u>One-Shot Game</u>	<u>Repeated Game</u>
Number of interactions	Once	Multiple times
Future consequences	None	Important
Cooperation	Less likely	More likely
Reputation	Not important	Very important
Punishment possible	No	Yes
Trust development	Difficult	Possible